

SYSTEM VERIFICATION REPORT

Automated Construction Management System: Nong Nai Chang Khon Foo

The primary objective of this verification document is to systematically prove that the "Nong Nai Chang Khon Foo" system functions correctly in alignment with the project requirements. Each test case evaluates critical workflow components, including the LINE Webhook connection, Gemini AI Entity Extraction, Data Validation (IF Logic), Google Sheets integration, and Gmail notification generation.

1. Test Case Table

No.	Test Case	Input (LINE Message)	Expected Result	Status
1	System Initialization & Greetings	"hi"	The system recognizes the input as a non-reporting interaction and prompts the user to submit a formal work report.	Pass
2	Nominal Reporting (Ahead of Schedule)	"โฟร์แมน ก โครงการ บ้านคุณ ค วันที่ 1 พ.ค. 2026 งานเสาเข็ม 50%"	The AI extracts the data, updates the Master Plan to "In Progress" (50%), and sends an email report to the PE noting the nominal/fast progress.	Pass
3	Delayed Reporting (Critical Path Impact)	"โฟร์แมน ก โครงการ บ้านคุณ ค วันที่ 7 พ.ค. 2026 งานเสาเข็ม 100% ผมจะเคลานได้วันที่ 8"	The AI detects a 2-day delay against the CPM (May 5), updates the status to "Done" (100%), and emails the PE recommending an Action Plan (e.g., Overtime).	Pass

No.	Test Case	Input (LINE Message)	Expected Result	Status
4	Incomplete Data (Validation Logic)	"โฟร์แมน โครงการ บ้านคุณ วันที่ พ.ค. งานเสียบ"	The IF Node intercepts the payload due to missing progress percentage and exact date. It replies via LINE to request the missing details.	Pass
5	Invalid Data / Gibberish	"dwdawdawfd"	The system rejects the unrecognized payload and replies via LINE requesting standard reporting parameters.	Pass

2. Evidence and Documentation

To substantiate the verification results, the following evidence is documented. All sensitive credentials (e.g., API Keys, Webhook URLs, OAuth Tokens) have been secured and obscured within the n8n Credentials Vault.

Evidence Source	Description
LINE Chat Interface Logs	Displays the full interaction flow between the Foreman and the Bot across all 5 test scenarios (Greetings, Nominal, Delayed, Incomplete, and Gibberish data).
Google Sheets Master Plan	Demonstrates the real-time database updates executed by the n8n Google Sheets Node, reflecting accurate percentage progress and status tracking.
Gmail API Action Plan Report	Showcases the automated emails received by the Project Engineer, featuring AI-generated Gap Analysis and strategic recommendations (e.g., Overtime).
n8n Execution Logs	Confirms the successful routing of data payloads through the Webhook, Edit Fields, AI Agent, and IF Function nodes without critical pipeline failures.

3. Test Result Explanation

Based on the simulated scenarios, the **Nong Nai Chang Khon Foo** system successfully fulfills its operational mandate. Initially, conversational logic was verified; the bot correctly differentiated between casual greetings and formal progress reporting.

In data processing tests, the Google Gemini 2.5 Flash AI Agent demonstrated high precision in **Entity Extraction**, accurately identifying project names, dates, and completion percentages from natural Thai language inputs. When presented with nominal data (Scenario 2), the pipeline flawlessly updated Google Sheets and notified the Project Engineer. Crucially, in Scenario 3 (Delayed Reporting), the AI performed **Predictive Analysis** against the Critical Path Method (CPM) baseline. It successfully recognized a 2-day delay and proactively formulated an Action Plan suggesting Overtime (OT) to mitigate cascading delays to subsequent predecessor tasks.

Furthermore, the **Validation Logic (IF Node)** performed perfectly. When presented with incomplete or gibberish inputs (Scenarios 4 and 5), the system intercepted the payload and instantly requested clarification from the user via LINE, thereby maintaining strict **Data Hygiene** within the central database.

4. Limitations Found

While the system performs exceptionally well, the following limitations were observed during testing:

Limitation	Explanation
Highly Ambiguous Thai Slang	While Gemini understands natural language well, highly localized construction slang or heavily misspelled terminology may occasionally cause the AI Agent to misclassify the entity, requiring manual correction.
Database Structural Rigidity	The Google Sheets Master Plan must retain its predefined column headers (e.g., Task Name, Start, End, Status). Modifying these headers manually will cause the AI's <code>update_master_plan</code> tool to fail.
API Rate Limits & Quotas	The system relies on Google Gemini API and n8n Webhook limits. Extremely high volumes of simultaneous reports from multiple site locations might trigger rate limiting if quotas are not adjusted for enterprise-scale deployment.

5. Verification Summary

The verification process confirms that the automated construction management system operates with high reliability. It successfully transforms the LINE application into an intelligent data ingestion point, effectively eliminating "Data Lag." The dual-layered architecture proves highly valuable: the IF Node enforces strict data hygiene, while the AI Agent acts as an automated **Strategic Consultant**. The system is fully functional, having successfully shifted the project management paradigm from a reactive reporting structure to a proactive, data-driven methodology.

Appendix: Evidence Screenshots



Figure A1: LINE Bot Interaction Logs across all 5 test scenarios

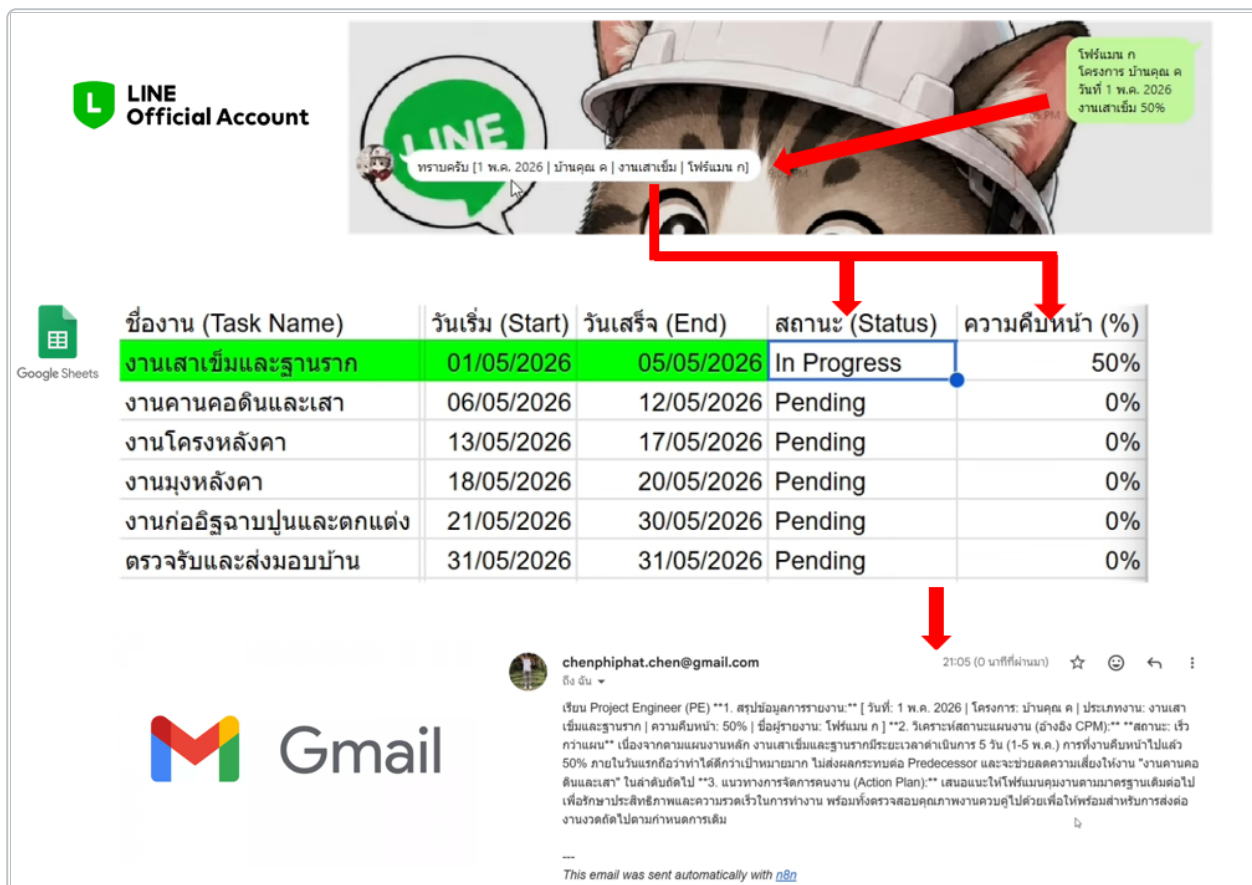


Figure A2: Database Update & Email Notification (Nominal Report)

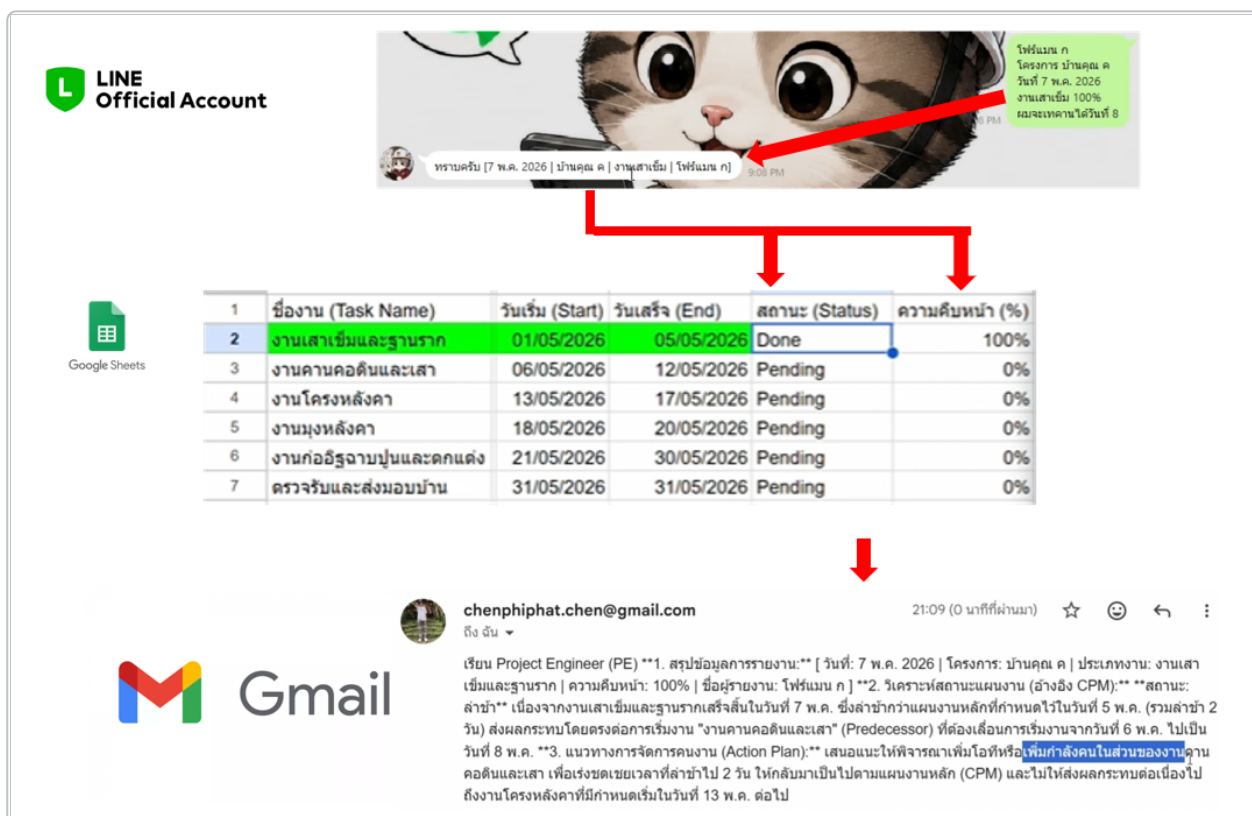


Figure A3: Database Update & Email Action Plan (Delayed Report)